

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10: SUB-STRAND 1.3 QUADRATIC EQUATIONS

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the sub-strand. Driving question: How can writing and solving a quadratic equation help us find unknown dimensions, quantities, and arrangements in real-life situations across Kenya? Think back to Achieng's chair problem at the Kisumu prize-giving ceremony and to your work in Lessons 1 to 8. Answer every question in the space under it. Show each step of your working, write your reasoning in full sentences, and give your final answers with units or a meaning where the question asks for one. Use r for the number of rows in Achieng's problem.

Part 1: Turning Achieng's seating problem into an equation

A school in Kisumu needs to arrange 120 chairs for guests in a rectangular block. The number of columns must exceed the number of rows by 7. (a) Write a Let statement for the number of rows and write an expression for the number of columns. (b) Write an equation that shows the total number of chairs. (c) Expand and rearrange it into the standard form $ax^2 + bx + c = 0$ and state the values of a , b and c . (d) Explain why Achieng could not solve this by simple division or guess-and-check, and why a squared term appears.

(a) Let r = the number of rows. Then the number of columns = $r + 7$. (b) Rows $\times$ columns = total chairs, so $r(r + 7) = 120$. (c) Expanding gives $r^2 + 7r = 120$. Subtracting 120 from both sides gives $r^2 + 7r - 120 = 0$. Here $a = 1$, $b = 7$ and $c = -120$. (d) Rows and columns are two linked unknowns whose product is known, so dividing 120 by one number does not give the answer, because the other number depends on the first. Guess-and-check gave her decimals or wrong totals. Both dimensions are written using the same letter r , so when we multiply them we get $r \times r = r^2$. This is why the equation is quadratic, with highest power 2, and not linear. [Mark: Let statement, product equation, correct standard form, correct a , b , c , and a reason linked to multiplying two expressions in the same unknown.]

Part 2: Solving by factorisation

Use the equation you formed, $r^2 + 7r - 120 = 0$. (a) Use the product-and-sum method: state the product and the sum you

(a) We need two numbers whose product is -120 (the constant term) and whose sum is $+7$ (the coefficient of r). The numbers are $+15$ and -8 , because $15 \times (-8) = -120$ and $15 + (-8) = 7$. (b) $(r + 15)(r - 8) = 0$. (c) If two factors multiply to give zero, at least one of them must be zero. So $r + 15 = 0$, which gives $r = -15$, or $r - 8 = 0$, which gives $r = 8$. The two roots are $r = 8$ and $r = -15$. (d) Expanding: $(r + 15)(r - 8) = r^2 - 8r + 15r - 120 = r^2 + 7r - 120$. This matches the original expression, so the factorisation is correct.

need, and name the two numbers that work. (b) Write the equation in factorised form. (c) Use the Zero Product Property to find both values of r. (d) Check your factorisation by expanding the brackets.	[Mark: correct product and sum, correct factors, both roots, and verification by expansion.]
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Part 3: Verifying and choosing the valid answer

(a) Substitute each of your two roots into the original equation $r^2 + 7r - 120 = 0$ and show what you get. (b) Which root can Achieng use for the chair arrangement? Give a reason based on the situation. (c) State the number of rows and the number of columns, and show that they give the correct total number of chairs.	(a) For $r = 8$: $8^2 + 7(8) - 120 = 64 + 56 - 120 = 0$, so it works. For $r = -15$: $(-15)^2 + 7(-15) - 120 = 225 - 105 - 120 = 0$, so it also works. Both roots satisfy the equation, so both are correct algebraic solutions. (b) Only $r = 8$ is valid. The number of rows is a count of real rows of chairs, so it cannot be negative, and $r = -15$ must be rejected. (c) There are 8 rows and $8 + 7 = 15$ columns. Check: $8 \times 15 = 120$ chairs, which matches the 120 guests.[Mark: substitution for both roots, rejection of -15 with a context reason, and correct rows, columns and product check.]
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Part 4: Using the method in a new Kenyan context

A market trader in Kisumu has a rectangular display table with an area of 28 m^2. The length of the table is 3 m more than its width. (a) Let w be the width in metres. Form a quadratic equation in standard form. (b) Solve it by factorisation and find both roots. (c) Decide which root is valid, using the context, and state the width and length of the table. (d) Show how you would verify your answer.	(a) Let w = the width in metres. Then the length = $w + 3$. Area: $w(w + 3) = 28$, so $w^2 + 3w = 28$, which gives $w^2 + 3w - 28 = 0$. (b) We need two numbers with product -28 and sum $+3$. They are $+7$ and -4. So $(w + 7)(w - 4) = 0$, which gives $w = -7$ or $w = 4$. (c) A width cannot be negative, so $w = -7$ is rejected and $w = 4$ is valid. The width is 4 m and the length is $4 + 3 = 7$ m. (d) Substitute $w = 4$ into the equation: $4^2 + 3(4) - 28 = 16 + 12 - 28 = 0$. Also, area = $4 \times 7 = 28 \text{ m}^2$, which matches the situation.[Mark: correct Let statement and equation, correct factorisation and both roots, rejection of the negative root with a reason, and verification.]
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Part 5: Answering the driving question

Using Achieng's chair problem and at least one other Kenyan	A quadratic equation helps whenever two related unknowns are multiplied to give a known total, such as the rows and columns of chairs in Kisumu or the width and length of a market table or a maize shamba. The pathway has these steps: (1) Understand the
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situation from the unit, explain how writing and solving a quadratic equation helps us find unknown dimensions, quantities or arrangements. Describe the full pathway from the real-life situation to the final answer, and explain why you must always check your answers in context.	situation and let a letter stand for one unknown, for example r = rows. (2) Write the other unknown as an expression, for example columns = $r + 7$. (3) Form the product equation $r(r + 7) = 120$ and rearrange it into standard form $r^2 + 7r - 120 = 0$. (4) Factorise by the product-and-sum method to get $(r + 15)(r - 8) = 0$. (5) Use the Zero Product Property to get both roots, $r = 8$ and $r = -15$. (6) Verify both roots by substitution, then use the context to decide which are valid. Algebra gives two correct roots, but the situation decides which one makes sense. Rows cannot be negative, so Achieng uses 8 rows and 15 columns, giving $8 \times 15 = 120$ chairs. In the same way, the trader's table has width 4 m and length 7 m. This shows that quadratic equations are a decision-making tool in design, farming, business and event planning across Kenya.[Mark: names the product-of-two-related-unknowns idea, gives the pathway in order, includes the context check, and uses a second Kenyan example.]
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Forming quadratic equations	Writes correct Let statements and expressions, forms the product equation, and rearranges it correctly into standard form with a, b and c identified, in every context.	Forms the equations correctly with minor slips in rearranging or in identifying a, b and c.	Has difficulty writing the expression for the second unknown or forms a linear equation or an equation not set to zero.
Solving by factorisation	Uses the product-and-sum method accurately, writes correct factors, applies the Zero Product Property to get both roots, and checks by expanding.	Factorises and finds both roots correctly but does not always check by expansion or explain the Zero Product Property.	Chooses the wrong factor pair, makes sign errors, or finds only one root.
Verification and contextual validity	Substitutes both roots, correctly rejects the invalid root with a clear reason from the context, and checks that the final dimensions give the known total.	Substitutes the roots and rejects the invalid root, but the reason or the final check is incomplete.	Does not verify by substitution or does not use the context to choose the valid answer.
Explaining the full journey and the driving question	Gives a clear, ordered pathway from real-life situation to verified answer, explains why the squared term appears, and links to at least two Kenyan contexts.	Describes most steps of the pathway with a Kenyan example but with limited explanation of why the squared term appears.	Lists some steps with little reasoning and few or no links to real-life situations.

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Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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